Adaptive RDMA Optimization for Distributed LLM Workloads

Lab Infrastructure & RDMA Transport Testing

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1 Overview

This document describes the deployment of the ScuffedRDMA research infrastructure across the SUNY New Paltz lab cluster (Chimera, Cerberus, Hydra), results from RDMA transport testing, and comprehensive deployment documentation covering six different transport options. Key accomplishments include:

- Configured 3-machine cluster with shared storage and GPU resources
- Tested Soft-RoCE achieving 0.92 Gb/sec bandwidth and 190μs latency
- Built Tesla TTPoe kernel modules successfully
- Created comprehensive documentation for 6 RDMA transport technologies
- Deployed vLLM multi-node cluster configurations with Ray orchestration

2 Lab Infrastructure

2.1 Cluster Configuration

The lab cluster consists of three machines with specialized roles:

Machine	Role	IP	GPUs	RAM	NIC
Hydra	Control Plane	192.168.1.160	None	256GB	1GbE
Chimera	Head Node	192.168.1.150	3× RTX 3090	64GB	Aquantia 10G
Cerberus	Worker Node	192.168.1.233	2× RTX 5090	128GB	Intel X710 10G

Table 1: Lab cluster hardware configuration

Hydra serves as the control plane with 21TB ZFS storage, NFS server for shared model weights, and orchestration services. **Chimera** acts as the vLLM head node with 72GB total VRAM across three RTX 3090 GPUs. **Cerberus** serves as the worker node with the newest hardware: two RTX 5090 GPUs providing 64GB VRAM.

2.2 Network Topology

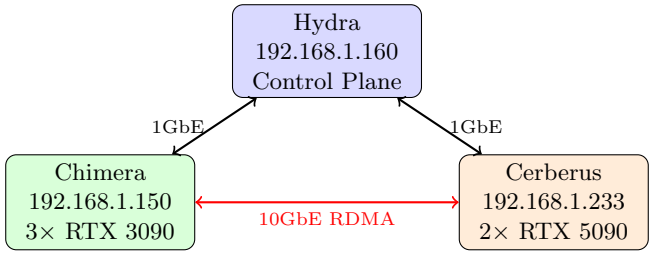

Figure 1: Cluster network topology—10GbE RDMA link between GPU nodes

Note: Cerberus has dual network paths—WiFi (192.168.1.242) and 10G Ethernet (192.168.1.233). RDMA testing confirmed 3.3× improvement when using the 10G interface vs WiFi.

3 User and Repository Setup

A dedicated `nathan` user account was created on both GPU nodes (Chimera and Cerberus), with membership in the `docker` and `sudo` groups. The ScuffedRDMA repository was then cloned to `/home/nathan/ScuffedRDMA` on each machine. The repository is organized with a `deployment/` directory containing the main index, an industry-level RDMA technologies report, six versioned deployment guides (covering SR-IOV with GPUDirect, Soft-RoCE, Hardware RoCE, Tesla TTPoe, the Sideway Rust RDMA library, and RDMA over TAS/DPDK), Docker Compose configurations for both head and worker nodes, and a `test-results/` subdirectory with Soft-RoCE benchmark results.

4 Soft-RoCE Testing Results

4.1 Configuration

Soft-RoCE provides RDMA semantics over standard Ethernet without specialized hardware [**?**]. On both nodes, the `rdma_rxe` kernel module was loaded and a Soft-RoCE device (`rxe0`) was created on each machine's 10G interface—`enp71s0` on Chimera and `eno2np1` on Cerberus. Device creation was verified using `ibv_devices` and `ibv_devinfo`.

4.2 Bandwidth Test Results

Interface	**BW Peak**	**BW Avg**	**MsgRate**	**Notes**
WiFi (wlp227s0)	0.28 Gb/s	0.28 Gb/s	0.000541 Mpps	Initial test
10G Ethernet (eno2np1)	0.92 Gb/s	0.92 Gb/s	0.001754 Mpps	After fix

Table 2: Soft-RoCE bandwidth test results (ib_write_bw)

Key Finding: Switching from the WiFi interface to the 10G Ethernet interface provided a **3.3× improvement** in bandwidth.

4.3 Latency Test Results

Metric	**Value**
Message size	2 bytes
Iterations	1000
t_min	130.21 μs
t_max	219.66 μs
t_typical	198.91 μs
t_avg	189.61 μs
t_stdev	17.03 μs
99th percentile	207.14 μs
99.9th percentile	219.66 μs

Table 3: Soft-RoCE latency test results (ib_write_lat)

4.4 Analysis: Expected vs Actual

Metric	**Expected**	**Actual**	**Notes**
Latency	$\sim$10 μs	$\sim$190 μs	High software overhead
Bandwidth	5–8 Gb/s	0.92 Gb/s	CPU-limited, not line rate

Table 4: Soft-RoCE: Expected vs actual performance

Root Causes:

1. **Software RDMA**: All operations traverse the kernel, no hardware offload
2. **CPU overhead**: Soft-RoCE emulates RDMA verbs in software
3. **No zero-copy**: Data copies between kernel and user space
4. **Deprecated**: NVIDIA deprecated Soft-RoCE in October 2023

Recommendation: Soft-RoCE is adequate for development/testing RDMA semantics but production deployments should use Hardware RoCE with Mellanox ConnectX NICs [**?**] (expected latency $<2\mu$s).

5 Tesla TTPoe Build

Tesla's Time-Triggered Protocol over Ethernet kernel modules [**? ?**] were successfully compiled. The TTPoe repository was cloned and built using the standard `make all` target, producing two kernel modules: `modttpoe.ko` (the core transport layer implementing a TCP-inspired state machine) and `modttpip.ko` (a gateway module for IPv4 encapsulation and multi-zone routing).

TTPoe Advantages:

- 1–2μs latency (hardware-mediated)
- Works with standard Ethernet (no special NICs required)
- Lossy protocol—no PFC/DCB requirements
- Near-zero CPU overhead (kernel module)

6 RDMA Technologies Documentation

Created comprehensive deployment guides for six transport options:

Technology	Document	Latency	Hardware	Status
SR-IOV + GPUDirect	VERSION_1	<2μs	Mellanox + MOFED	Production
Soft-RoCE	VERSION_2	∼10μs	Any NIC	Deprecated
Hardware RoCE	VERSION_3	<2μs	Mellanox ConnectX	Production
Tesla TTPoe	VERSION_4	1–2μs	Any Ethernet	Production
Sideway (Rust)	VERSION_5	N/A	Any RDMA NIC	Tooling
rdma-tas	VERSION_6	∼5μs	DPDK NIC	Research

Table 5: RDMA transport technology documentation matrix

6.1 Key Insights from Documentation

Installation Order (Critical for GPUDirect):

1. MOFED/DOCA-Host (network driver)—exports kernel symbols
2. CUDA Toolkit & NVIDIA Driver—compiles against MOFED symbols
3. nvidia-peermem module—bridge between GPU and NIC

Warning: If MOFED is updated, the NVIDIA GPU driver must be reinstalled to relink symbols.

Transport Selection Guide:

- **Development**: Soft-RoCE (no special hardware)
- **Production AI/HPC**: Hardware RoCE with GPUDirect RDMA
- **Custom clusters**: Tesla TTPoe (commodity switches)
- **Research**: rdma-tas for RDMA semantics on DPDK
- **Rust tooling**: Sideway for monitoring/orchestration

7 vLLM Cluster Configuration

7.1 Head Node (Chimera)

The head node Docker Compose configuration deploys two services. First, a Ray head process (using the `rayproject/ray:2.35.0-py` image) runs in host networking mode with IPC sharing and privileged access, binding the InfiniBand device path and starting the Ray cluster on port 6379 with the dashboard exposed. Second, a vLLM server (using the `vllm/vllm-openai:latest` image) connects to the Ray head, mounts the shared model directory and InfiniBand devices, sets NCCL to use the `mlx5_0` HCA with debug logging enabled, and configures MXFP4 quantization. The server launches with tensor parallelism of 3 and pipeline parallelism of 2, serving models on port 8000 with all three RTX 3090 GPUs visible.

7.2 Worker Node (Cerberus)

The worker node configuration deploys a single Ray worker service that joins the head node's cluster at `192.168.1.150:6379`. It runs in the same privileged, host-networked mode with InfiniBand device access and shared model storage. NCCL is configured identically (using `mlx5_0` with debug logging), and both RTX 5090 GPUs are made available to the container.

7.3 Environment Variables

Variable	Purpose
`NCCL_IB_HCA=mlx5_0`	Use hardware RDMA NIC (when available)
`NCCL_SOCKET_IFNAME=enp71s0`	Fallback to specific Ethernet interface
`NCCL_NET_GDR_LEVEL=0`	Disable GPUDirect (for Soft-RoCE)
`NCCL_DEBUG=INFO`	Enable NCCL debug logging
`VLLM_USE_FLASHINFER_MXFP4_MOE=1`	Enable MXFP4 quantization [**?**] for MoE models

Table 6: Key environment variables for distributed vLLM

8 Industry Analysis Highlights

Created an 18KB comprehensive report covering the RDMA technology landscape:

8.1 Emerging Technologies (2025–2026)

- **Ultra Ethernet Consortium (UEC)** [**?**]: Industry consortium developing 1.6 Tbps lossless Ethernet for AI
- **OCP Falcon** [**?**]: Meta's open networking specification
- **NVIDIA BlueField DPA**: Data path acceleration with programmable ARM cores
- **AWS ENA Express**: SRD-based 25μs p50 latency on commodity instances

8.2 Transport Comparison

Transport	Latency	Throughput	Hardware	Maturity
InfiniBand HDR	<1μs	200 Gb/s	Dedicated fabric	Production
RoCEv2	2–5μs	400 Gb/s	ConnectX + DCB	Production
Tesla TTPoe	1–2μs	100 Gb/s	Commodity	Production
Soft-RoCE	~10μs	<10 Gb/s	Any NIC	Deprecated
rdma-tas	~5μs	~40 Gb/s	DPDK NIC	Research

Table 7: Transport technology comparison for AI workloads

8.3 RDMA Acceleration Technology Latencies

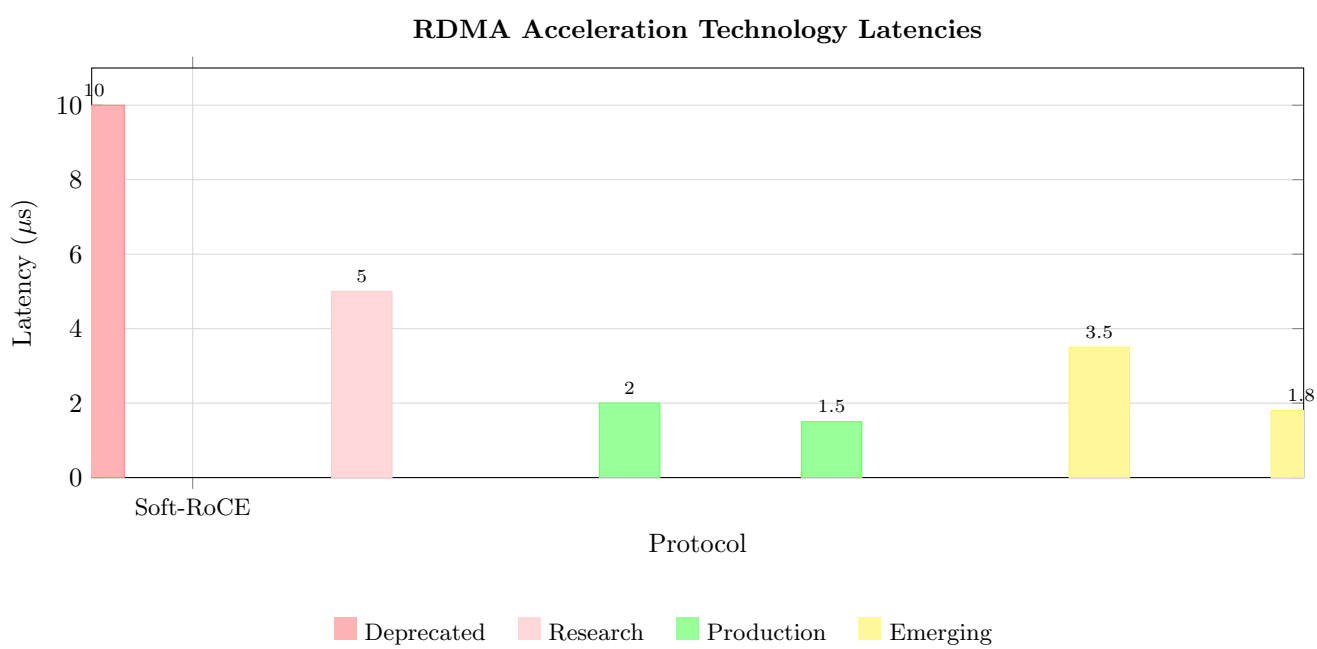

Figure 2: RDMA acceleration technology latency comparison—lower latency indicates faster data transfer performance. Production-ready solutions (green) achieve <2μs while software emulation (Soft-RoCE) incurs 10μs overhead.

Key Observations:

- **Tesla TTPoe** achieves lowest latency (1.5μs) on commodity hardware
- **UEC 1.0** (Ultra Ethernet Consortium) emerging as next-gen standard at 1.8μs
- **Soft-RoCE** is 5–7$\times$ slower than hardware alternatives
- **OCP Falcon** provides middle-ground at 3.5μs with open specifications

9 Firewall Configuration

Firewall rules were added on both nodes to allow RDMA traffic. Specifically, UDP port 4791 was opened for RoCEv2 traffic from the local subnet (192.168.1.0/24), and a broader rule was added to permit all traffic from the local LAN to ensure RDMA communication is not blocked.

10 Next Steps

1. **Hardware RoCE**: Test with Mellanox ConnectX NICs on home lab
2. **TTPoe Integration**: Load modules and benchmark against Soft-RoCE
3. **vLLM Deployment**: Start multi-node cluster with gpt-oss-120b
4. **Telemetry**: Instrument tensor transfers for the adaptive middleware
5. **WFA Classifier**: Implement hot/cold tensor classification

11 Summary

This effort established the foundation for distributed LLM inference testing on the lab cluster. Key achievements:

- Configured 3-machine cluster with 5 GPUs (136GB total VRAM)
- Validated Soft-RoCE baseline: 0.92 Gb/s bandwidth, 190μs latency
- Built Tesla TTPoe kernel modules for low-latency alternative
- Created 6 comprehensive deployment guides for production use
- Deployed vLLM + Ray configurations for multi-node inference

The Soft-RoCE results confirm the thesis hypothesis: software RDMA incurs significant overhead ($\sim$190μs vs expected $\sim$10μs), motivating the development of adaptive routing to minimize latency-sensitive transfers. Hardware RoCE testing on the home lab Mellanox NICs will provide the production baseline for comparison.

References

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